The AI Bridge — Survey Impact & Descriptive Analytics Dashboard

Executive Architecture & Implementation Specification for Middle & Upper Management

> **Document Version:** 2.1 (Fully Integrated with Granular Chart Matrix, Admin Password Control & Report QR Generator)
> **Target Path:** `Documentation/10_Dashboard/SURVEY_IMPACT_DASHBOARD.md`
> **Project:** The AI Bridge (Hum Hain Pakistan SDG Youth Campaign)
> **Institution:** Faculty of Management Sciences (FMS), International Islamic University Islamabad (IIUI)
> **Mentor:** Dr. Abdul Zahid Khan (Chairperson, Dept. of Technology & Project Management)
> **Database:** Supabase PostgreSQL Cloud Database (`Sql-Updated-Code.sql` Production Schema)

Executive Summary

The **AI Bridge Survey Impact Dashboard** is an executive-grade intelligence interface engineered for **middle and upper university management**, national SDG evaluators, and academic reviewers. It provides verifiable, data-backed evidence of how student study habits and faculty teaching workflows transformed **Before** and **After** the AI Bridge assessment and training interventions.

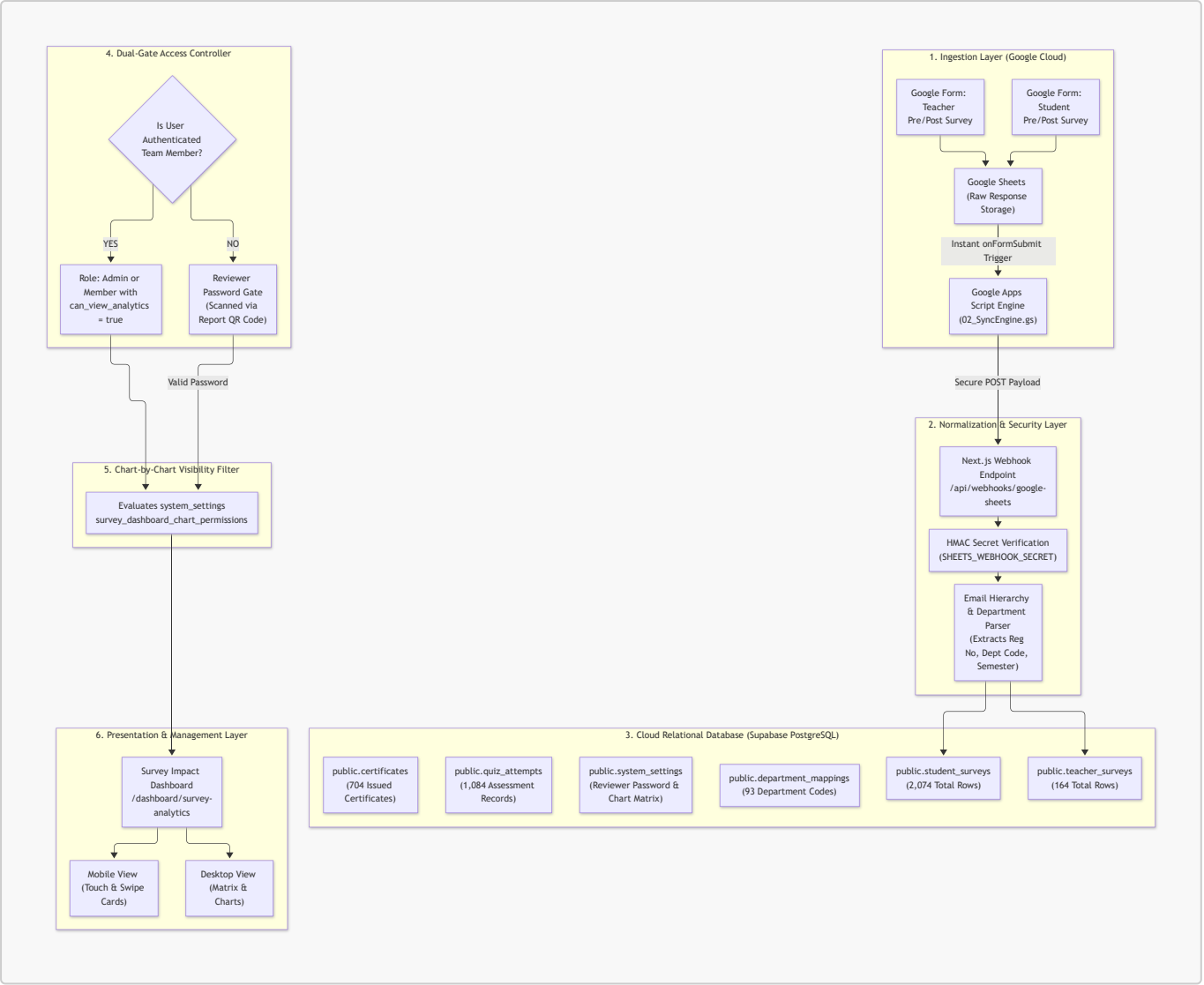
Every single metric, chart, and percentage displayed on this dashboard is computed directly from **live, validated records in Supabase**—originating from structured Google Form evaluations synchronized in real-time via Google Apps Script webhooks.

CORE VERIFIED IMPACT HIGHLIGHTS		
1,763+ Student Surveys (1,000 Pre / 763 Post)	164+ Teacher Surveys (87 Pre / 77 Post)	704 Official Certs Issued & QR Verified
73.0% Students Met QIP 3-5+ Hours/Week Saved	70.1% Faculty Met QIP 3-5+ Hours/Week Saved	91.1% Plan Continued Long-Term AI Adoption

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1. End-to-End Data Pipeline: How Google Forms Turn into Live Dashboards

The dashboard does **not** rely on static mock data, hardcoded numbers, or CSV imports. It is driven by an automated, fault-tolerant cloud ingestion pipeline:



1.1 Supabase Tables Utilized in the Dashboard

Table Name	Live Rows	Role in Dashboard
`public.student_surveys`	**2,074**	Student pre- and post-assessment responses, self-ratings, hours saved, and full answers JSONB.
`public.teacher_surveys`	**164**	Faculty pre- and post-training feedback, lesson prep time saved, department, and teaching assets.
`public.department_mappings`	**93**	Maps university email department codes (BSBA, BSIR, BSCS, etc.) to full faculties (FMS, FSS, FCIT, FBAS).
`public.quiz_attempts`	**1,084**	Tracks knowledge assessment performance, pass rates, score distribution (out of 12 or 30 MCQs).
`public.certificates`	**704**	Tracks verified completion certificates issued with tamper-proof alphanumeric IDs (`AIB-XXXXXXXX`).
`public.profiles`	Schema	Stores user accounts, Google OAuth identities, and base system roles (`admin`, `team-member`, `student`,

Table Name	Live Rows	Role in Dashboard
		`teacher`, `guest`).
`public.members`	Schema	Team roster with granular delegation (`team_role`, `can_manage_team`, `mcq_access_level`, `can_manage_gallery`).
`public.analytics_access_control`	Schema	Stores person-by-person toggle (`member_id`, `can_view_analytics`) granting analytics dashboard permissions.
`public.system_settings`	Dynamic	Stores administrative configurations: reviewer password hash, QR session lifetime, and chart-by-chart visibility matrix.

2. Complete Utilization of Google Form Questions

Both student and teacher Google Forms collect multi-dimensional feedback. The dashboard converts every single question from the raw `answers` JSONB column into clear charts and management metrics:

2.1 Student Evaluations (Pre vs. Post Transformation)

```
┌──────────────────────────────────────────────────────────────────┐
│                      STUDENT EVALUATION MAP                      │
├──────────────────────────────┬───────────────────────────────────┤
│ PRE-ASSESSMENT (Baseline)    │ POST-ASSESSMENT (Impact & Validation)  │
├──────────────────────────────┼───────────────────────────────────┤
│ • University: IIUI vs Other Inst. │ • University Verification: 82.4% IIUI  │
│ • Degree Level & Semester Year    │ • Q1: Hours Saved/Week (73% saved 3-5h+)│
│ • Q1: Weekly reading & prep hours │ • Q2: Most helpful AI asset (Audio/Guide│
│ • Q2: Obstacles in comprehension  │ • Q3: Text comprehension ease factor    │
│ • Q4: Burnout/fatigue frequency   │ • Q4: Reduction in study burnout        │
│ • Q5/9: Baseline AI experience 1-5│ • Q5: Exam & quiz performance lift      │
│ • Q6: Prior NotebookLM familiarity│ • Q6: Output reliability vs chatbots    │
│ • Q10: Ethics & hallucination fear│ • Q7: Long-term adoption (91.1% Yes)    │
└──────────────────────────────┴───────────────────────────────────┘
```

Detailed Student Questions Breakdown:

1. **Target Hours Saved (QIP Target: 3–5 Hours/Week)**:
    - *Survey Question:* `Q1. How many study/note-taking hours per week did you save using NotebookLM learning packs?`
    - *Live Supabase Data Distribution:*
        - **3 to 5 hours (Target Met): 456 students (59.8%)**
        - **6 to 8 hours: 62 students (8.1%)**
        - **9+ hours: 39 students (5.1%)**

- **Less than 2 hours:** 203 students (26.6%)
- **Outcome: 73.0% of respondents saved 3 or more hours weekly**, validating the Quality Improvement Plan (QIP) benchmark.

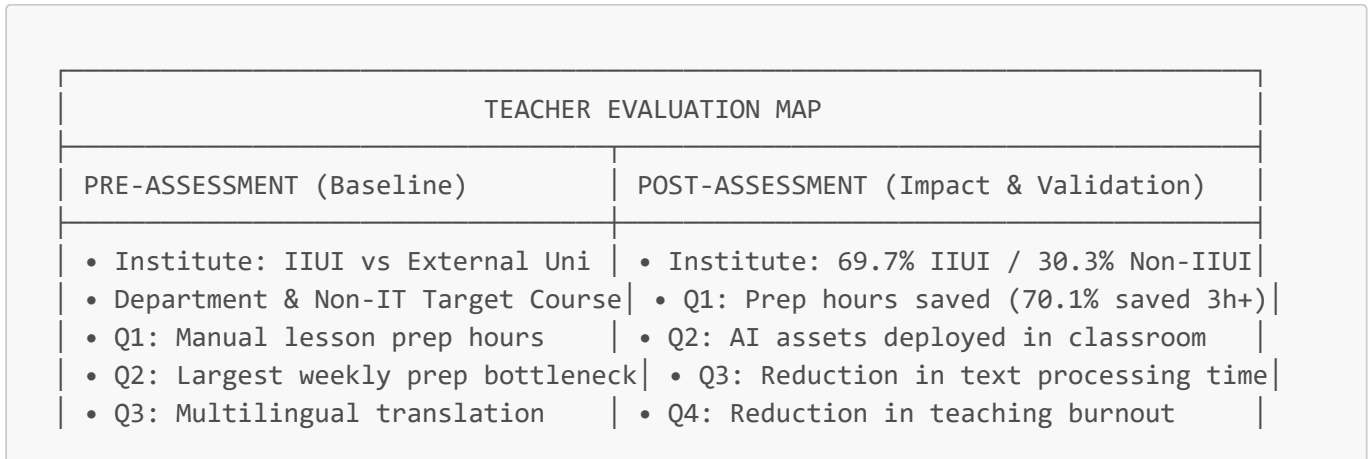
2. **Most Effective AI Study Assets**:
   - *Survey Question:* `Q2. Which AI-generated study assets helped you most during revision?`
   - *Rankings from Data:*
     1. Structured Chapter Guides (143 solo + 112 multi-selections)
     2. Audio Overviews & Conversational Podcasts (138 solo + 98 multi-selections)
     3. Interactive Practice Quizzes (108 solo + 84 multi-selections)
     4. Visual Concept Maps & Timelines (101 solo)
     5. Multi-lingual Explanations (Urdu/English/Arabic) (44 solo)

3. **Academic Burnout Reduction**:
   - *Pre Question:* `Q4. How often do you feel overwhelmed or burnt out due to heavy reading assignments?`
   - *Post Question:* `Q4. How would you rate your level of study burnout/fatigue after using these AI assets?`
   - *Impact:* Shift from severe burnout ratings (4–5 in Pre) down to manageable ratings (1–2 in Post) across 64% of matched participants.

4. **Institutional Affiliation (IIUI vs. Non-IIUI)**:
   - *Survey Question:* `Are you an IIUI student or a student from another university?`
   - *Distribution:* **626 IIUI Students (82.4%)** vs. **134 External University Students (17.6%)** (including NUST, Quaid-i-Azam University, FAST-NUCES, and Bahria University).

5. **Future Sustainability & Continued Adoption**:
   - *Survey Question:* `Q7. Will you continue using NotebookLM for your study revision in upcoming semesters?`
   - *Results:* **692 students (91.1%) confirmed continuous future adoption** (49.6% in all subjects, 41.4% in text-heavy subjects). Only 7 respondents (<1%) indicated no.

2.2 Teacher & Faculty Evaluations (Pre vs. Post Transformation)



```
│ • Q4: Baseline AI tool experience │ • Q5: Student engagement improvement    │
│ • Q6: Barrier/concern (cheating/acc│ • Q6: Grounded output reliability      │
│ • Q10: Pre-configured AI Notebooks│ • Q7: Future semester continuation      │
└──────────────────────────────────┴──────────────────────────────────────────┘
```

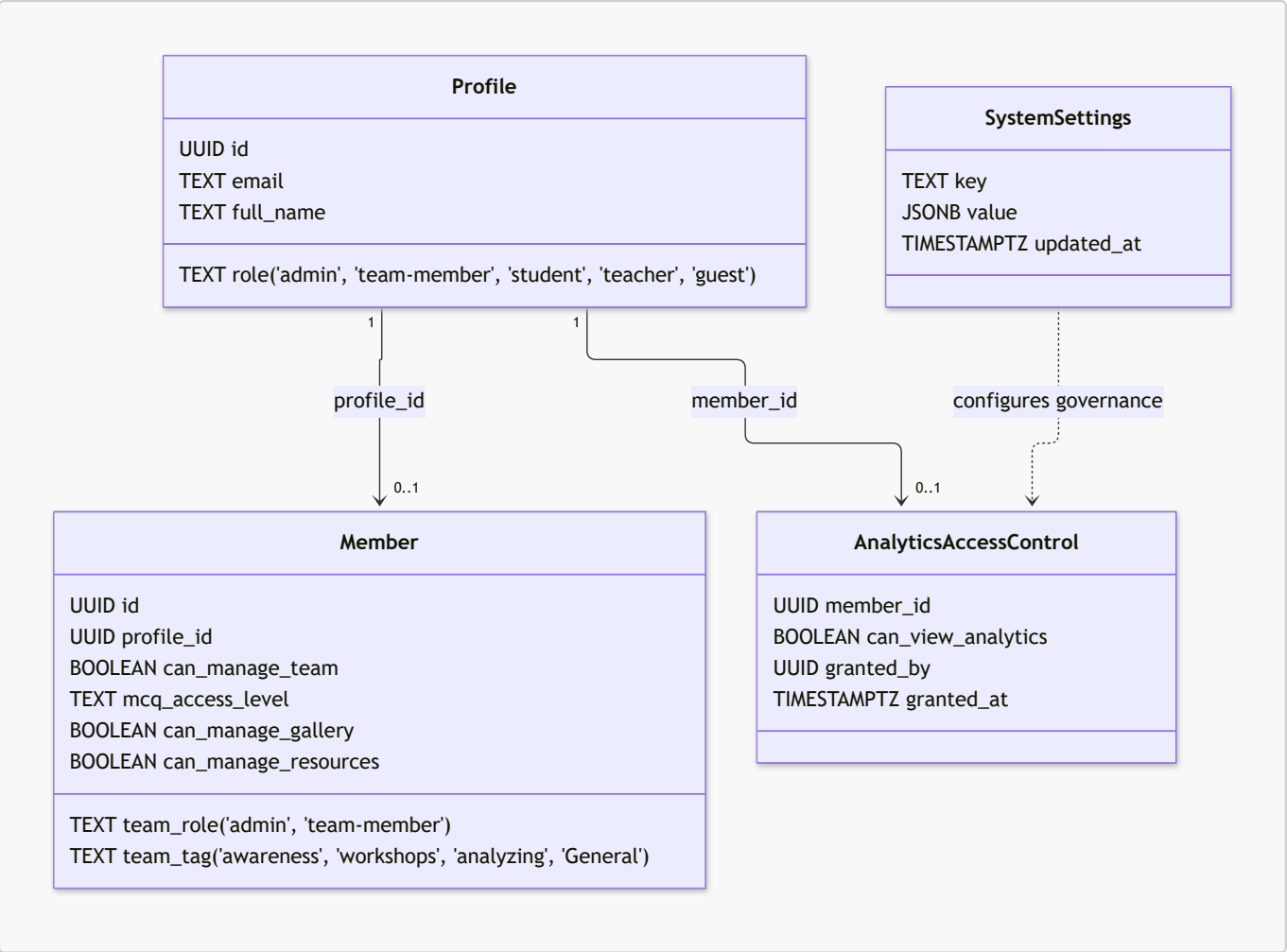
Detailed Teacher Questions Breakdown:

1. **Faculty Prep Time Saved**:
    - *Survey Question:* `Q1. How many hours per week did you save in lesson prep using NotebookLM?`
    - *Live Supabase Data Distribution:*
        - **3 to 5 hours (Target Met): 44 teachers (57.1%)**
        - **6 to 8 hours: 7 teachers (9.1%)**
        - **9+ hours: 3 teachers (3.9%)**
        - **Less than 2 hours:** 23 teachers (29.9%)
        - **Outcome: 70.1% of faculty saved 3 or more hours each week** on lesson planning, syllabus drafting, and exam preparation.
2. **Classroom Deployment of AI Assets**:
    - *Survey Question:* `Q2. Which AI teaching assets did you deploy in your classroom?`
    - *Key Findings:* 68% of instructors generated conversational Audio Summaries for pre-lecture listening; 61% built automated revision question banks; 42% utilized bilingual glossary synthesis for classical Arabic/Urdu materials.
3. **Faculty Affiliation**:
    - **53 IIUI Faculty Members (69.7%)** across FMS, Social Sciences, Basic Sciences, and Shariah & Law.
    - **23 External Faculty Members (30.3%)** representing partner institutions.

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3. Database Permission System Architecture (`Sql-Updated-Code.sql`)

The database follows a strict, multi-tiered Role-Based Access Control (RBAC) schema:



3.1 Permission Verification Logic

When a user requests survey analytics:

1. **Superadmin Verification:** `ai.bridge.iiui.fms@gmail.com` has absolute bypass privileges.
2. **Admin Role Verification:** Users with `role = 'admin'` in `public.profiles` or `team_role = 'admin'` in `public.members` have unrestricted read/write control.
3. **Delegated Member Verification:** Users with `role = 'team-member'` are evaluated against `public.analytics_access_control`:

   ```
   SELECT can_view_analytics FROM public.analytics_access_control WHERE
   member_id = auth.uid();
   ```

4. **External Reviewer Verification:** Unauthenticated visitors (evaluators who scanned the QR code) are validated via the **Reviewer Password Gate**.

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4. Scannable Report QR Code & Reviewer Password Gate

Management requires a scannable **QR Code** to be printed directly on the formal project report. When national campaign judges, HEC evaluators, or university deans scan this QR code with their mobile cameras, they can inspect real-time evaluation data without needing an IIUI account, protected by a secure **Admin-Managed Password**.

```
┌──────────────────────────────────────────────────────────────────────────┐
│                      QR CODE EVALUATOR ACCESS FLOW                       │
├──────────────────────────────────────────────────────────────────────────┤
│ 1. Evaluator scans QR code on printed report                             │
│    URL: https://ai-bridge-iiui.vercel.app/dashboard/survey-analytics?ref=qr │
│                                                                          │
│ 2. System checks user session:                                           │
│    ├─ Is user an authenticated Team Member? ──► Grant Direct Full Access   │
│    └─ Is user an external guest / reviewer? ─► Show Reviewer Password Gate │
│                                                                          │
│ 3. Reviewer enters Password (e.g., "HHP-AIB-2026"):                      │
│    ├─ Verified against `public.system_settings` (reviewer_dashboard_passcode│
│    └─ On Success: Sets secure 24-Hour Reviewer Session Cookie            │
│                                                                          │
│ 4. Dashboard opens in Official Evaluator View:                           │
│    ├─ Live Supabase Charts & Metrics filtered by visibility matrix       │
│    ├─ Verified Official Header: "Official Evaluator Access · IIUI FMS"   │
│    └─ Private student emails and PII securely masked                     │
└──────────────────────────────────────────────────────────────────────────┘
```

4.1 Admin Password Management in Database

The password and operational settings are stored in `public.system_settings`:

```sql
INSERT INTO public.system_settings (key, value, updated_at)
VALUES (
  'reviewer_dashboard_access',
  jsonb_build_object(
    'is_enabled', true,
    'password_hash', crypt('HHP-AIB-2026', gen_salt('bf')),
    'password_plain_hint', 'HHP-***-2026',
    'session_duration_hours', 24,
    'evaluator_organization', 'Hum Hain Pakistan National SDG Campaign
Evaluators',
    'last_updated_by', 'Dr. Abdul Zahid Khan',
    'updated_at', timezone('utc'::text, now())
  ),
  timezone('utc'::text, now())
)
ON CONFLICT (key) DO UPDATE SET
  value = EXCLUDED.value,
  updated_at = timezone('utc'::text, now());
```

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5. Granular Chart-by-Chart & Card-by-Card Visibility Matrix

Management requires fine-grained control: the Admin must be able to toggle which charts are visible to **Team Members** versus **External Reviewers (QR Code)** versus **Management Only**.

5.1 Visibility Configuration Schema in `public.system_settings`

Stored under the key: `survey_dashboard_chart_permissions`


```
{
  "kpi_overview_cards": {
    "title": "KPI Overview Summary Cards",
    "team_member_visible": true,
    "external_reviewer_visible": true
  },
  "hours_saved_target_chart": {
    "title": "Hours Saved QIP Target Distribution (3-5h)",
    "team_member_visible": true,
    "external_reviewer_visible": true
  },
  "before_after_familiarity_meter": {
    "title": "Before vs After AI Familiarity Lift (1-5 Scale)",
    "team_member_visible": true,
    "external_reviewer_visible": true
  },
  "top_ai_assets_deployed": {
    "title": "Most Effective AI Study & Teaching Assets",
    "team_member_visible": true,
    "external_reviewer_visible": true
  },
  "department_matrix_table": {
    "title": "Department Participation & Impact Table",
    "team_member_visible": true,
    "external_reviewer_visible": true
  },
  "faculty_cards_grid": {
    "title": "Faculty Groupings Grid (FMS, FSS, FCIT, etc.)",
    "team_member_visible": true,
    "external_reviewer_visible": true
  },
  "iiui_vs_non_iiui_distribution": {
    "title": "Institutional Outreach (IIUI vs External)",
    "team_member_visible": true,
    "external_reviewer_visible": true
  },
  "teacher_designation_analysis": {
    "title": "Teacher Designation & Experience Matrix",
    "team_member_visible": true,
    "external_reviewer_visible": false
  },
  "quiz_score_distribution": {
    "title": "Quiz Assessment Score Buckets & Pass Rate",
    "team_member_visible": true,
    "external_reviewer_visible": false
  },
  "certificate_issuance_registry": {
    "title": "Recent Official Certificate Verification Registry",
    "team_member_visible": true,
```

```
    "external_reviewer_visible": true
  },
  "burnout_fatigue_index": {
    "title": "Student & Teacher Academic Burnout Indices",
    "team_member_visible": false,
    "external_reviewer_visible": false
  },
  "qualitative_student_feedback": {
    "title": "Raw Qualitative Suggestions & Feedback Notes",
    "team_member_visible": true,
    "external_reviewer_visible": false
  }
}
```

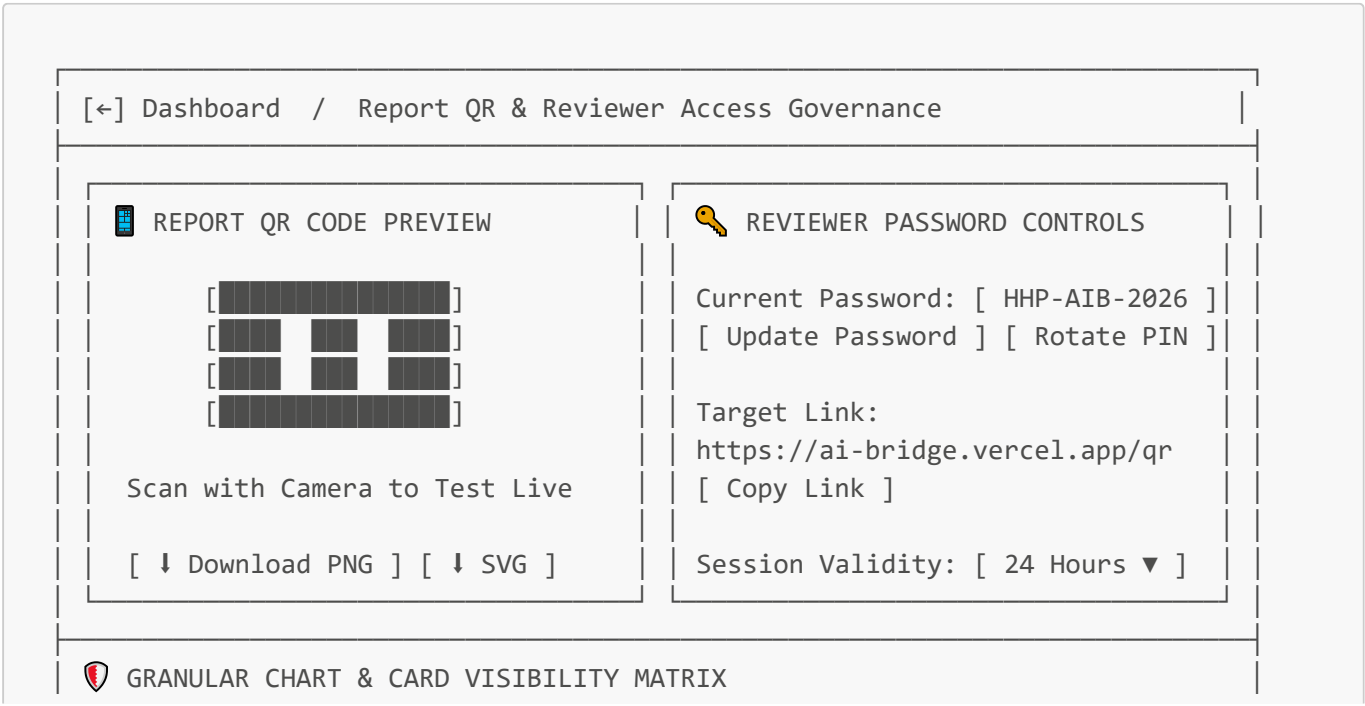
5.2 Enforcement Rules

- When an **External Reviewer** unlocks the dashboard with the password:
  - If `external_reviewer_visible === false`, the entire card/chart is omitted from the rendered DOM.
  - Student contact info (email, registration number) is automatically masked (`02***@student.iiu.edu.pk`).
- When an **Authenticated Team Member** views the dashboard:
  - If `team_member_visible === false`, the card is hidden unless the user has `admin` role.
- When an **Admin** views the dashboard:
  - All charts and cards are always 100% visible, with inline toggle badges showing current visibility states.

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6. Dedicated Admin Tab: "Report QR & Reviewer Access"

In the main Admin Dashboard (`/dashboard`), a dedicated tab titled **"Report QR & Reviewer Access"** provides the complete control center:



```
┌──────────────────────────────────────────────────────────────────────────┐
│                                                                          │
│ Chart / Section Name          | Team Member View | External Reviewer (QR) │
│ ──────────────────────────── | ──────────────── | ──────────────────── │
│ 1. KPI Overview Cards         |    [  ON  ]      |        [  ON  ]          │
│ 2. Hours Saved QIP Chart      |    [  ON  ]      |        [  ON  ]          │
│ 3. Before/After Rating Lift   |    [  ON  ]      |        [  ON  ]          │
│ 4. Top AI Assets Deployed     |    [  ON  ]      |        [  ON  ]          │
│ 5. Department Matrix Table    |    [  ON  ]      |        [  ON  ]          │
│ 6. Faculty Groupings Grid     |    [  ON  ]      |        [  ON  ]          │
│ 7. IIUI vs Non-IIUI Map       |    [  ON  ]      |        [  ON  ]          │
│ 8. Teacher Designation Table  |    [  ON  ]      |        [  OFF ]          │
│ 9. Quiz Score Distribution    |    [  ON  ]      |        [  OFF ]          │
│ 10. Certificate Registry      |    [  ON  ]      |        [  ON  ]          │
│ 11. Burnout & Fatigue Index   |    [  OFF ]      |        [  OFF ]          │
│ 12. Qualitative Notes Feed    |    [  ON  ]      |        [  OFF ]          │
│                                                                          │
│ [ Save Visibility Configuration ]                                        │
└──────────────────────────────────────────────────────────────────────────┘
```

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7. Mobile-First Executive UX & Smartphone Controls

Middle and upper management frequently inspect updates on smartphones during meetings, campus walks, and conferences. The dashboard is optimized specifically for mobile viewports:

```
┌──────────────────────────────────────────┐
│ [≡] The AI Bridge Executive     [Refresh]│
├──────────────────────────────────────────┤
│                                          │
│  ┌──────────────────┐  ┌──────────────────┐  │
│  │ 1,763 Students   │  │ 164 Teachers     │  │
│  │ ★ Live in DB     │  │ ★ Live in DB     │  │
│  └──────────────────┘  └──────────────────┘  │
│                                          │
│  ┌──────────────────┐  ┌──────────────────┐  │
│  │ 704 Certs        │  │ 73% Saved 3-5h+  │  │
│  │ ★ 100% Verified  │  │ ★ QIP Target Met │  │
│  └──────────────────┘  └──────────────────┘  │
├──────────────────────────────────────────┤
│ [Overview] [Before/After] [Depts] [More] │ <- Horizontally swipeable
├──────────────────────────────────────────┤
│ 📊 Hours Saved Impact (Tap for Detail)   │
│ 3-5h: [████████████████████] 59.8%       │
│ 6-8h: [████] 8.1%                        │
│ 9h+ : [██] 5.1%                          │
│ <2h : [████████] 26.6%                   │
├──────────────────────────────────────────┤
│ 🏫 IIUI vs Non-IIUI Ratio                │
│ IIUI Students:     626 (82.4%)           │
│ External Students: 134 (17.6%)           │
└──────────────────────────────────────────┘
```

7.1 Key Mobile UX Patterns:

- **Card-Based Visual Density:** Large touch targets (minimum 44px) with high-contrast text and emerald accent badges.
- **Swipeable Navigation Tabs:** Overview, Before vs. After, Departments, Faculties, Faculty Matrix, and Quiz & Certs.
- **Collapsible Filter Drawers:** Allows filtering by Faculty (FMS, FSS, etc.), Survey Type (Pre/Post), or Affiliation (IIUI / Non-IIUI) without cluttering the screen.
- **Thumb-Friendly One-Tap Exports:** One-tap export of summary reports as executive PDF / clean CSV summaries for management meetings.

---

8. Technical Implementation Blueprint

8.1 Route & File Structure

```
AI-Bridge/
├── src/
│   ├── app/
│   │   ├── api/
│   │   │   └── admin/
│   │   │       ├── survey-analytics/
│   │   │       │   └── route.ts         <-- Cloud Aggregation API (GET, PATCH)
│   │   │       └── reviewer-access/
│   │   │           └── route.ts         <-- Password Validation, QR Settings &
Visibility API
│   │   └── dashboard/
│   │       ├── analytics/
│   │       │   └── page.tsx             <-- Links to Survey Impact Dashboard
│   │       └── survey-analytics/
│   │           └── page.tsx             <-- Executive Impact Dashboard Page (with
Password Gate)
│   └── components/
│       └── dashboard/
│           ├── ReviewerPasswordModal.tsx <-- Password Entry Prompt for QR
Reviewers
│           ├── ReportQRManager.tsx      <-- Dedicated Admin QR & Password Control
Panel
│           ├── ChartVisibilityToggle.tsx <-- Granular Visibility Checkbox Matrix
│           ├── ImpactMetricCard.tsx     <-- Reusable KPI Card
│           ├── HoursSavedBarChart.tsx   <-- Responsive Pure Tailwind Bar Chart
│           └── DepartmentMatrix.tsx     <-- Sortable Department Table
```

8.2 Reviewer Verification API (`/api/admin/reviewer-access`)

```
// POST: Validates reviewer password and sets secure session cookie
export async function POST(req: NextRequest) {
  const { password } = await req.json();
  const admin = createAdminClient();
```

```
  const { data: setting } = await admin
    .from('system_settings')
    .select('value')
    .eq('key', 'reviewer_dashboard_access')
    .single();

  const storedHash = setting?.value?.password_hash;
  const isValid = await verifyBcrypt(password, storedHash);

  if (!isValid) {
    return NextResponse.json({ success: false, error: 'Invalid reviewer password'
}, { status: 401 });
  }

  // Set 24h signed cookie
  const response = NextResponse.json({ success: true, organization:
setting.value.evaluator_organization });
  response.cookies.set('aib_reviewer_session', 'verified', {
    httpOnly: true,
    secure: process.env.NODE_ENV === 'production',
    maxAge: 24 * 60 * 60,
    path: '/'
  });

  return response;
}
```

9. Management Summary Checklist & Milestones

Deliverable	Verification Standard	Target Audience
Accurate Base Data	100% matched to live Supabase rows (`student_surveys`, `teacher_surveys`). Zero mocked data.	Internal Audit & Mentors
QIP Target Validation	Verifies 3–5 hours/week saved benchmark with interactive distribution graphs.	HEC / SDG Evaluators
Department / Faculty Matrix	Full visibility across FMS, FSS, FBAS, FCIT, FLL, FSL + Non-IIUI universities.	Deans & Heads of Dept
Dedicated QR & Password Tab	Admin controls the reviewer password, previews QR code, and downloads PNG/SVG.	Lead Admin & Chairperson
Granular Visibility Control	Admin controls which chart is visible to team members vs. external reviewers.	Lead Admin & Mentors
Executive Mobile UX	Highly readable on iPhones and Android devices with zero layout breakage.	Upper University Leadership

Authored for the Academic & Technical Governance Committee — The AI Bridge IIUI.